Quiz — 1.4.3 Boolean Algebra — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (8 marks — 1 each) 1. **D** — AND outputs 1 only when both inputs are 1. 2. **C** — XOR outputs 1 when inputs differ. 3. **B** — NAND and NOR are the universal gates. 4. **B** — `¬(A + B) = ¬A · ¬B` (break the bar, swap OR→AND). 5. **C** — double negation. 6. **C** — half adder Sum = A ⊕ B. 7. **B** — D-type flip-flop stores one bit on a clock edge. 8. **C** — K-map groups are powers of two and as large as possible.

Section B

9. [5] Q = ¬(A·B) + (A·C):

A	B	C	A·B	¬(A·B)	A·C	Q
0	0	0	0	1	0	**1**
0	0	1	0	1	0	**1**
0	1	0	0	1	0	**1**
0	1	1	0	1	0	**1**
1	0	0	0	1	0	**1**
1	0	1	0	1	1	**1**
1	1	0	1	0	0	**0**
1	1	1	1	0	1	**1**

Mark: A·B column correct (1); ¬(A·B) column correct (1); A·C column correct (1); Q correct for rows 1–4 (1); Q correct for rows 5–8 — note only row A=1,B=1,C=0 gives Q=0 (1).

10. [3] Simplify `¬(¬A + B)`: - Apply De Morgan's to the outer bar: `¬(¬A + B) = ¬(¬A) · ¬B` (break the bar, change + to ·, negate each term). (1) - Apply double negation: `¬(¬A) = A`. (1) - Result: **`A · ¬B`**. (1)

11. [3] Half adder with A = 1, B = 1: - **Sum = A ⊕ B = 1 ⊕ 1 = 0** (1) - **Carry = A · B = 1 · 1 = 1** (1) - Correct expressions stated (Sum = A⊕B, Carry = A·B) (1). (So 1 + 1 = binary `10`: Sum 0, Carry 1.)

12. [3] Half adder truth table:

A	B	Sum (A⊕B)	Carry (A·B)
0	0	**0**	**0**
0	1	**1**	**0**
1	0	**1**	**0**
1	1	**0**	**1**

Mark: Sum column correct (1); Carry column correct (1); fully correct table with 1+1 giving Sum 0 Carry 1 (1).

13. [3] `A · (A + B)`: - This is the **Absorption law**: `A · (A + B) = A`. (1 for naming Absorption) - Result: `A`. (1) - Justification (1): expand `A·(A+B) = A·A + A·B = A + A·B` (idempotent A·A = A); then `A + A·B = A` (A absorbs A·B), so the whole expression reduces to A.

Section C

14. [5] (a) **[1] Sum = A ⊕ B ⊕ Cin**.

(b) **[2] Cout = (A · B) + (Cin · (A ⊕ B))**. (1 mark for the A·B term, 1 mark for the Cin·(A⊕B) term ORed together. Equivalent correct forms such as `A·B + B·Cin + A·Cin` also gain full marks.)

(c) **[2]** A **half adder** adds only **two** bits (A and B) and has **no carry-in**, so it cannot accept a carry from a previous column (1). A **full adder** adds **three** bits (A, B and Cin), so full adders can be **chained** (a ripple-carry adder) with each carry-out feeding the next column's carry-in, allowing multi-bit binary numbers to be added correctly (1).

Total: 8 + 17 + 5 = 30 marks.